



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

NATIONAL SENIOR CERTIFICATE

GRADE 12

ENGINEERING GRAPHICS AND DESIGN P1

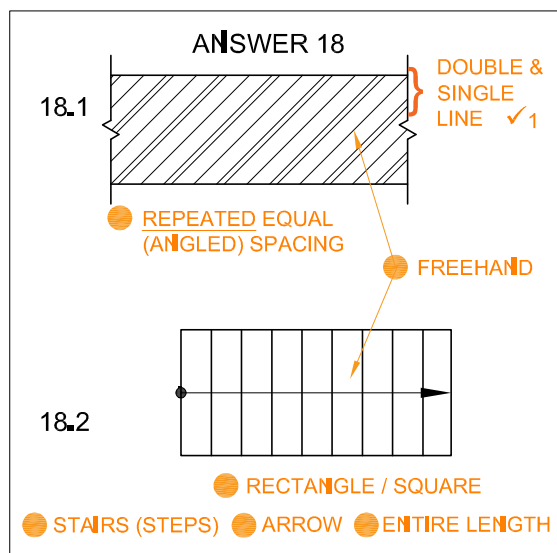
NOVEMBER 2022

MARKING GUIDELINES

MARKS: 100

These marking guidelines consist of 8 pages.

ANSWERS		
1	PROJECT NUMBER	1
2	MR N KEENAN	1
3	SITE PLAN	1
4	PRINTING	1
5	LISA B	1
6	15000	1
7	ARCHITECT and CLIENT / MR N KEENAN	2
8	2	1
9	RAINWATER DOWN-PIPE	1
10	E / SOUTH / TOP LEFT	1
11	4,57	1
12	NEW ✓ IRON / STEEL ✓	2
13	NORTH POINT	1
14	SWING / OPENING OF THE GATES	1
15	WATT METER / ELECTRICAL METER	1
16	FALL / GRADIENT / SLOPE / DROP	1
17	SOUTH WEST / SW	2
18	See below	4
19		3
20		3
	TOTAL	30



ANSWER 19
Show ALL calculations. ✓
APPLYING CORRECT FORMULA ✓

$$P = AB + BC + CD + DE$$

$$= AB + BC + CD + DE (- \text{GATE})$$

$$= (24,23 + 8,37 + 18,38 + 31,2) - 3,6$$

$$= 78,58 \quad \text{ANSWER IN METRES} \quad \checkmark$$

ANSWER 20
Show ALL calculations. ✓
APPLYING CORRECT FORMULA ✓

$$A = L \times B$$

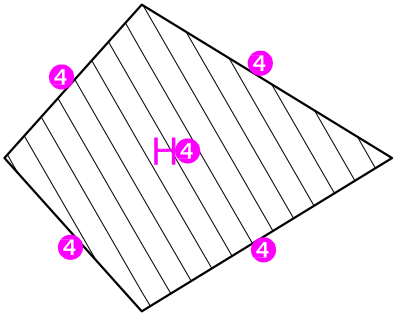
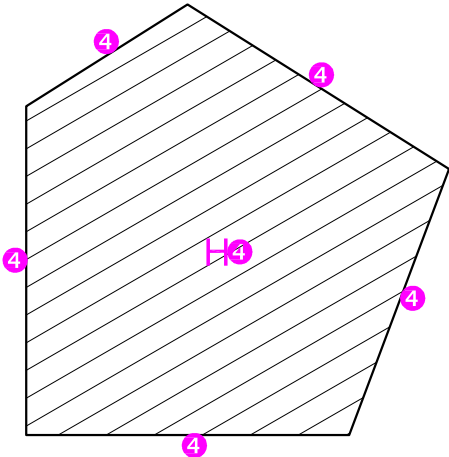
$$= (10,8 \times 6) + (13,2 \times 6) + (6,7 \times 6,7)$$

$$= 64,8 + 79,2 + 44,89$$

$$= 188,89 \quad \text{m}^2 \quad \checkmark$$

ANSWER IN SQUARE METRES AND m² SHOWN ✓

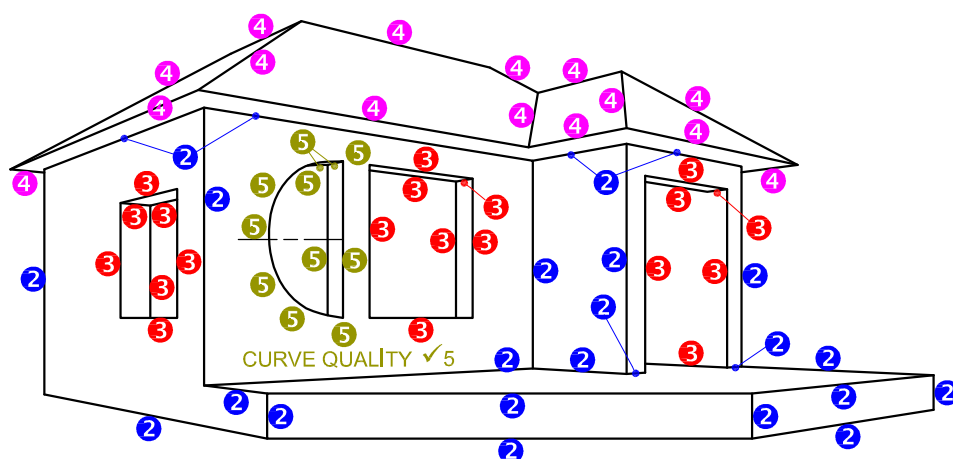
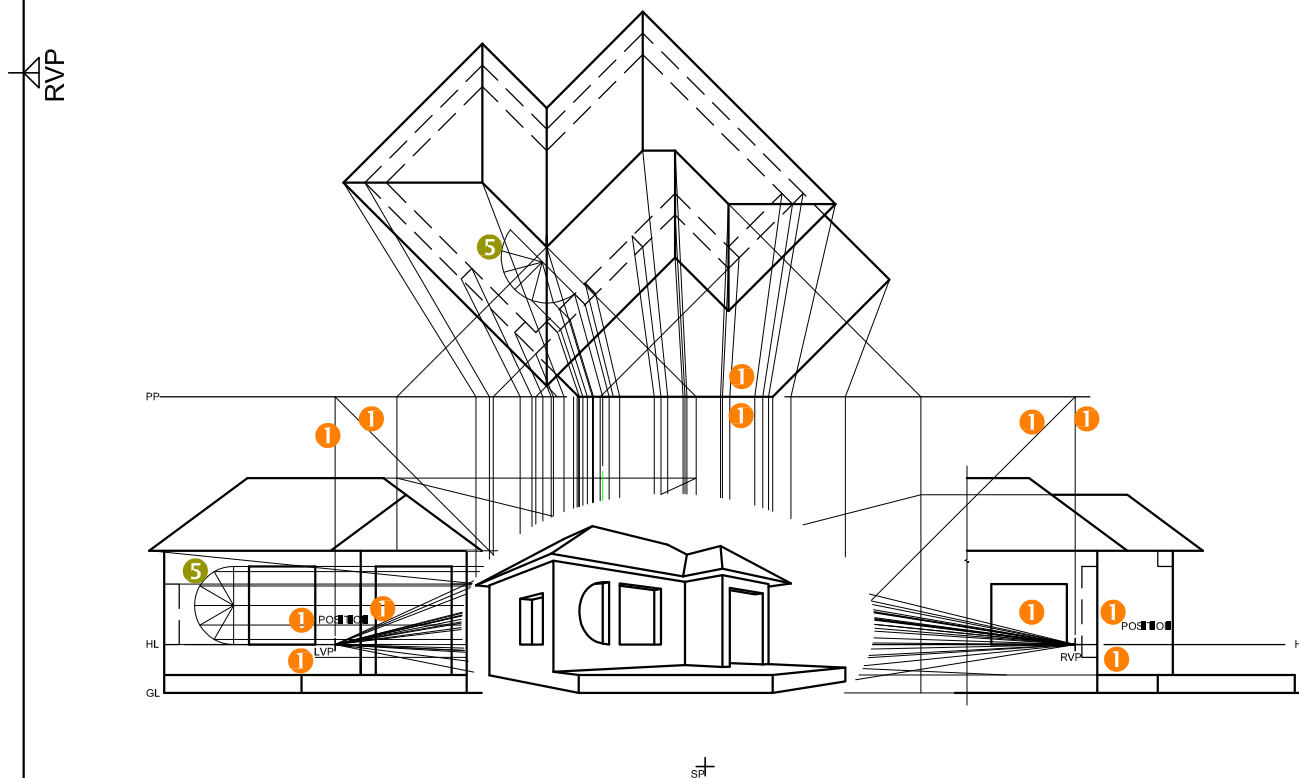
TRUE SHAPE
ANY RECOGNISED METHOD ✓ 4



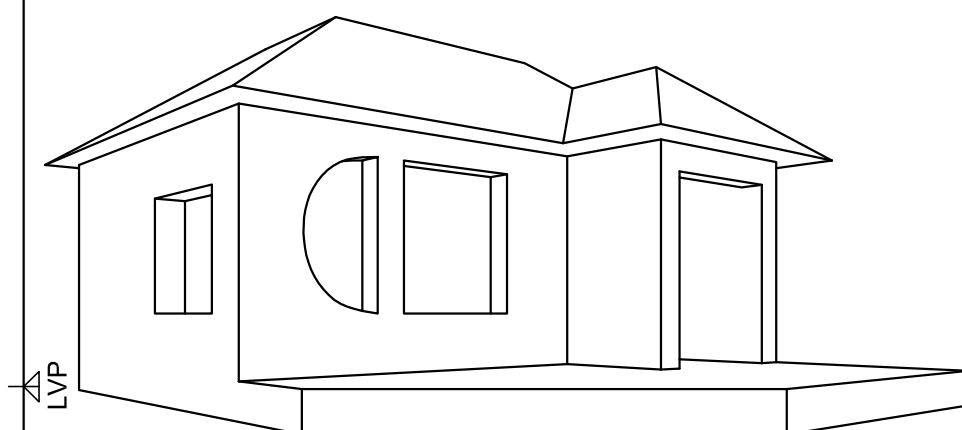
5. NOTE ON CORRECT HATCHING
DIFFERENT DIRECTIONS ✓ 5

ASSESSMENT CRITERIA		
4	TRUE SHAPE	6 ¹ / ₂
5	CORRECT HATCHING	1
SUBTOTAL		7 ¹ / ₂

PAPER 1 QUESTION 2
GRADE 12
NOVEMBER 2022
MARKING GUIDELINES

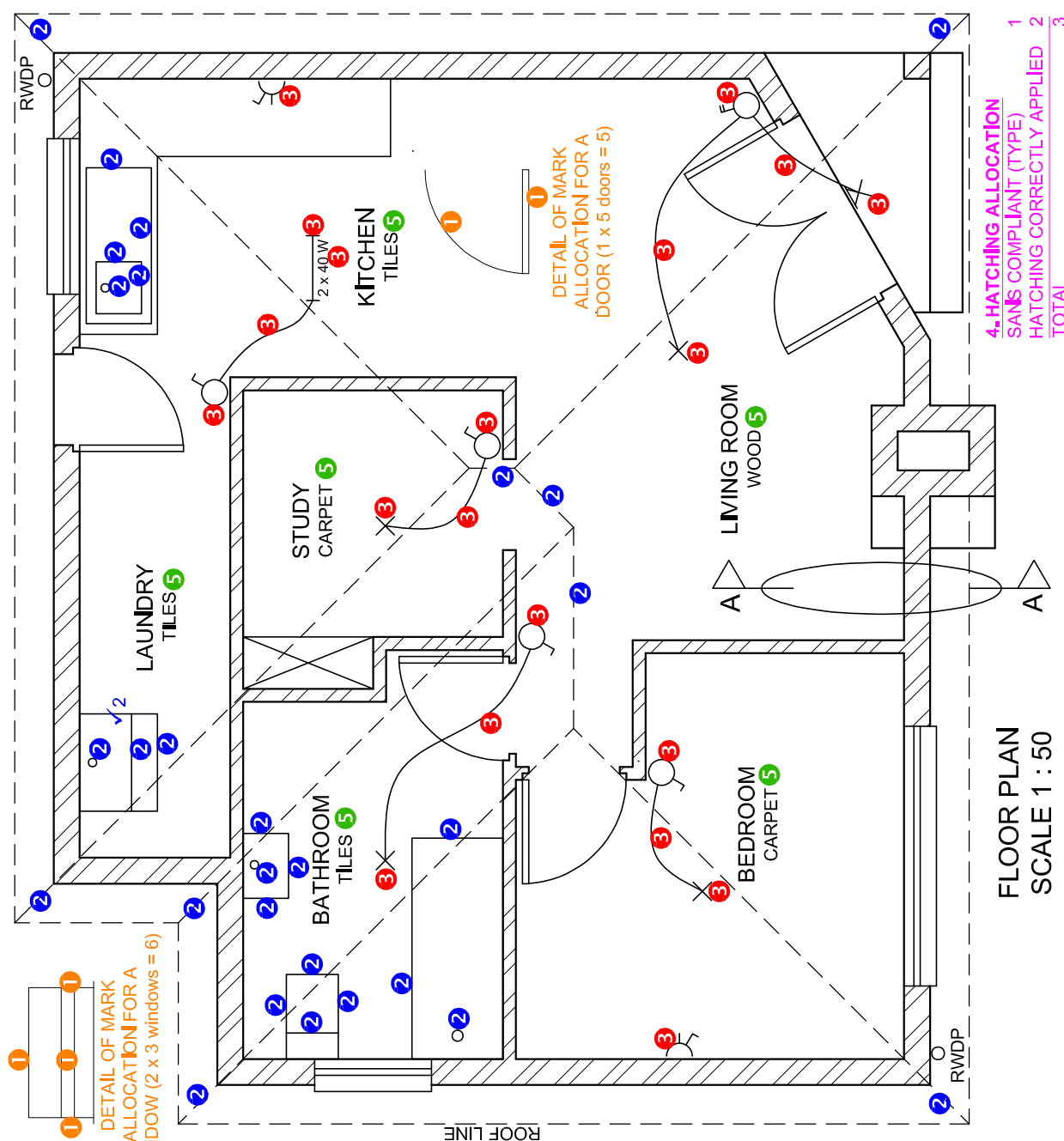


SP

**ASSESSMENT CRITERIA**

1	CONSTRUCTION	6
2	WALL + BASE	10 $\frac{1}{2}$
3	WINDOWS + OPENING	10
4	ROOF	7 $\frac{1}{2}$
5	ARC	7
TOTAL		41

PAPER 1 QUESTION 3
 GRADE 12
 NOVEMBER 2022
 MARKING GUIDELINES



ASSESSMENT CRITERIA	
FLOOR PLAN	
	POSSIBLE
1	DOORS + WINDOWS
2	FIXTURES + ROOF LINES
3	ELECTRICAL
4	HATCHING
5	LABELS
SUBTOTAL	
42	

PAPER 1 QUESTION 4
GRADE 12
NOVEMBER 2022
MARKING GUIDELINE

HEIGHT @ 60 mm

GL

FFL

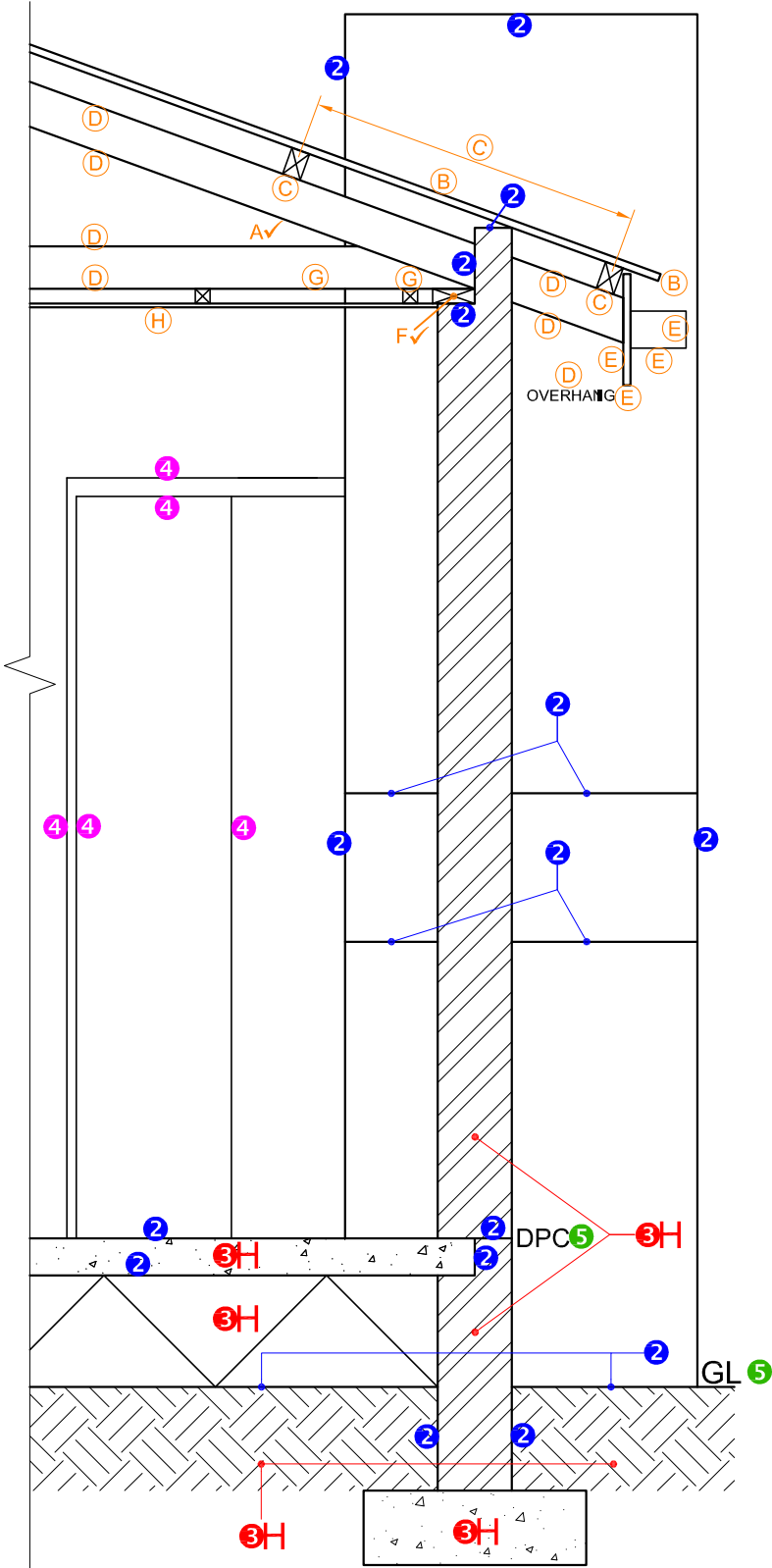
SOUTH ELEVATION

ASSESSMENT CRITERIA	
SOUTH ELEVATION	
1	ROOF + RWDP 8 $\frac{1}{2}$
2	WALLS + STEP + CHIMNEY + FFL 7
3	DOOR + WINDOW 7
4	LABELS 1
SUBTOTAL 23 $\frac{1}{2}$	

1. MARK ALLOCATION FOR ROOF

A: ROOF ANGLE	1
B: ROOF SHEET	1
C: PURLIN + SPACING	1,5
D: TRUSS + OVERHANG	3,5
E: FASCIA BOARD + GUTTER	2
F: WALL PLATE	1
G: BRANDING + SPACING	1
H: CEILING BOARD	0,5
TOTAL	11,5

3. HATCHING FOR ROUGH TIMBER	
SANS COMPLIANT (TYPE)	0,5
HATCHING CORRECTLY APPLIED	
APPLIED	0,5
TOTAL	1



ASSESSMENT CRITERIA		
DETAILED SECTION		
1	ROOF DETAIL	11 1/2
2	SLAB + WALL + CHIMNEY	8
3	HATCHING	3 1/2
4	DOOR	2 1/2
5	LABELS	1
SUBTOTAL		26 1/2